رَبِّ اشْرَحْ لِي سَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِلن لِسَانِي يَفْقَهُوا قَوْلِي

Rabbish Rahli Sadri wa Yassirli Amri Wahlul Uqadatam Millisani Yafkahu Qawli.

My Lord, Uplift my heart for me, make my task easy, and remove the dullness from my tongue so that they may understand me.
(Surah Ta Ha: 25-26-27-28)

ABSORPTION AND FACTORS AFFECTING ABSORPTION OF DRUGS

DR SANA IRSHAD

PHARMACOLOGY

OBJECTIVES

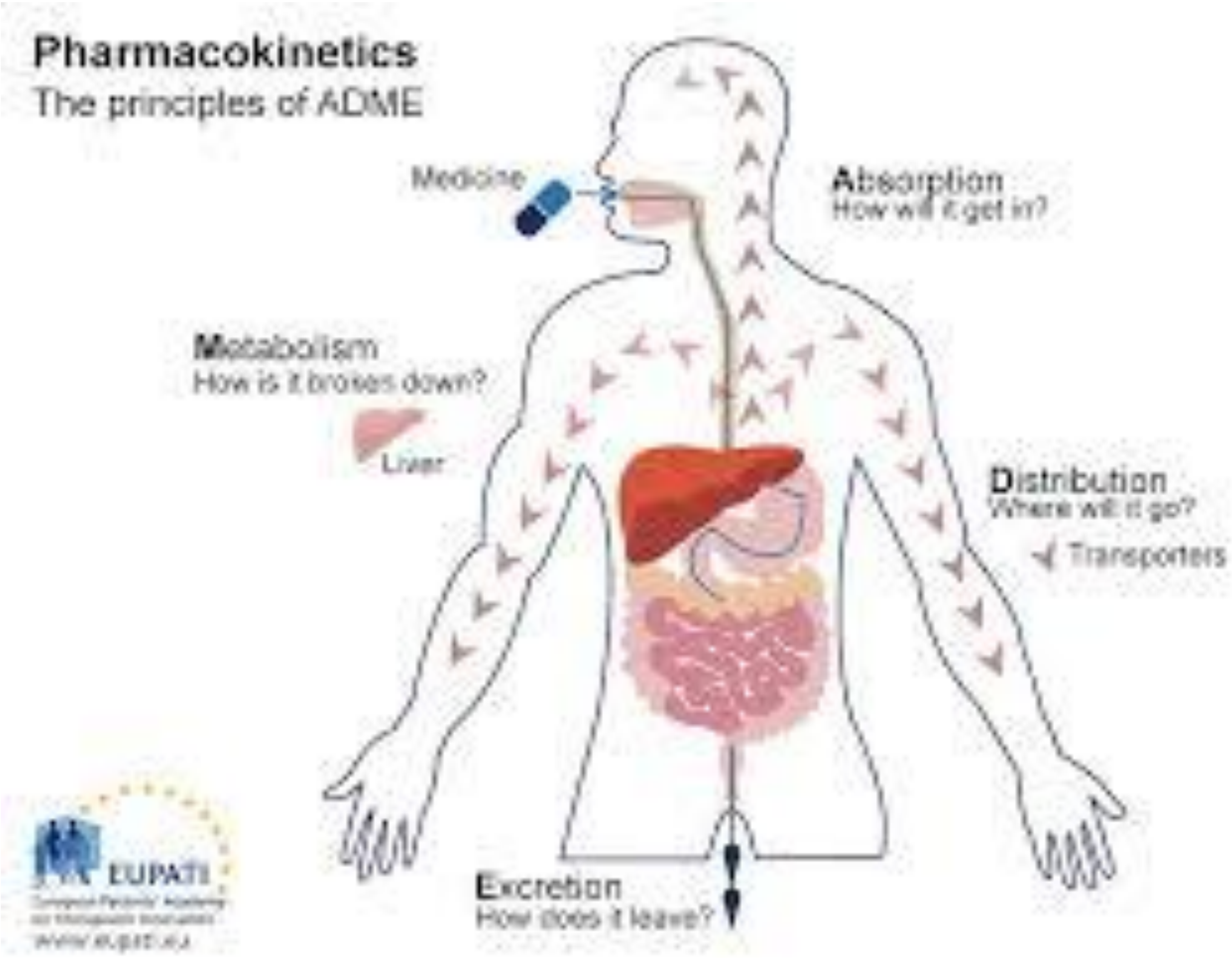
By the end of the lecture the students of BDS 1st year will be able to:

- Define absorption.
- Discuss the factors affecting drug absorption.
- Define bioavailability.
- Define first pass effect.

Pharmacokinetics (PK)

- *Action of body on the drug.*

- *Pharmacokinetics includes*
- **Absorption of drug**
- **Distribution**
- **Metabolism**
- **Excretion**


Pharmacokinetics
The principles of ADME
Medicine
Absorption
How will it get in?
Metabolism
How is it broken down?
Liver
Distribution
Where will it go?
Transporters
Excretion
How does it leave?
EUPATI
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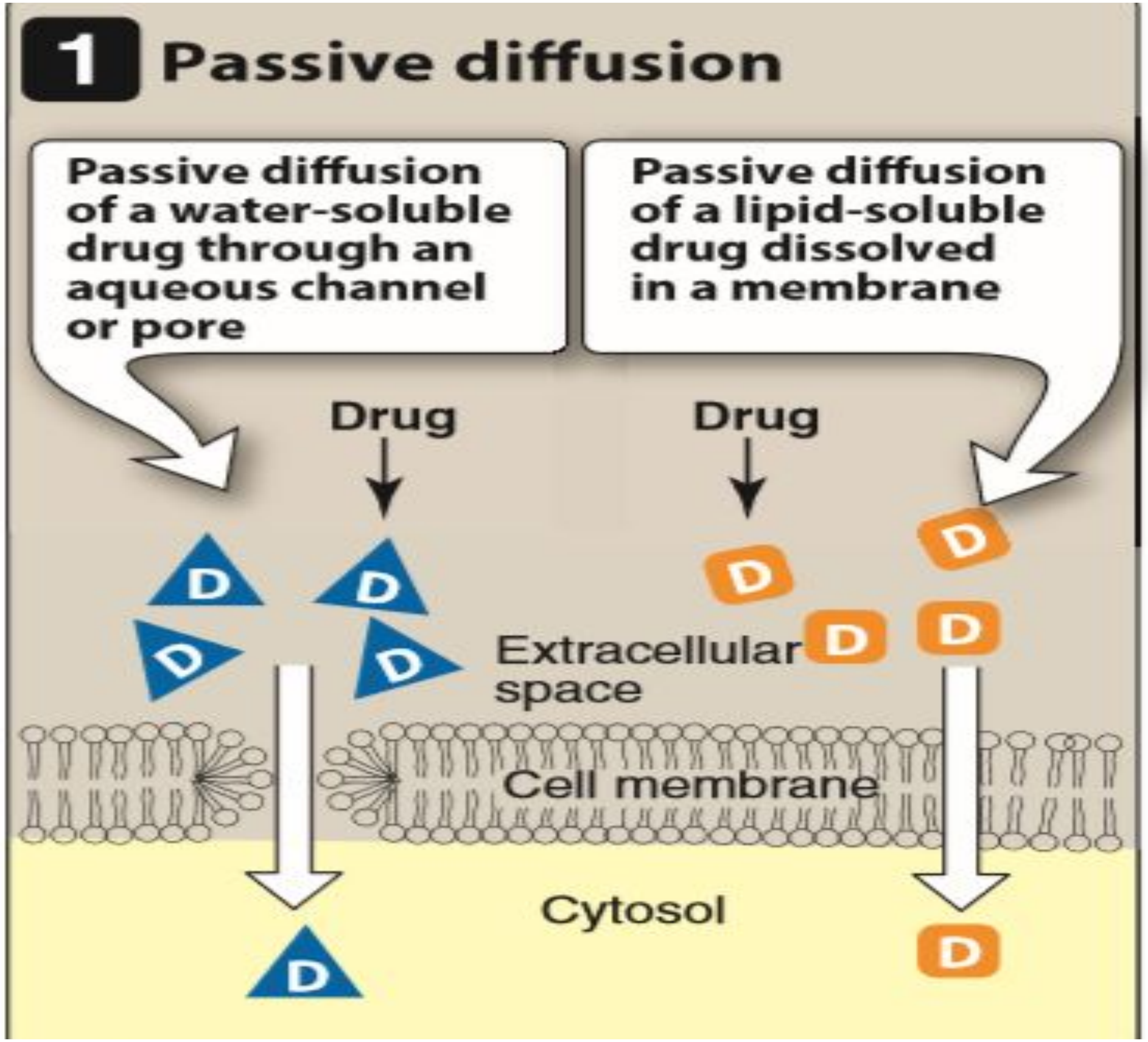
ABSORPTION

Absorption is the transfer of a drug from the site of administration to the bloodstream.

Mechanisms of absorption of drugs from the GI tract

1. ***Passive diffusion:***

- The drug moves from a region of high concentration to one of lower concentration.
- Concentration gradient across a membrane.
- It does not involve a carrier, is not saturable.
- Vast majority of drugs are absorbed by this mechanism.
- Water-soluble drugs penetrate the cell membrane through aqueous channels or pores
- Whereas lipid-soluble drugs readily move across most biologic membranes due to their solubility in the membrane lipid bilayers.

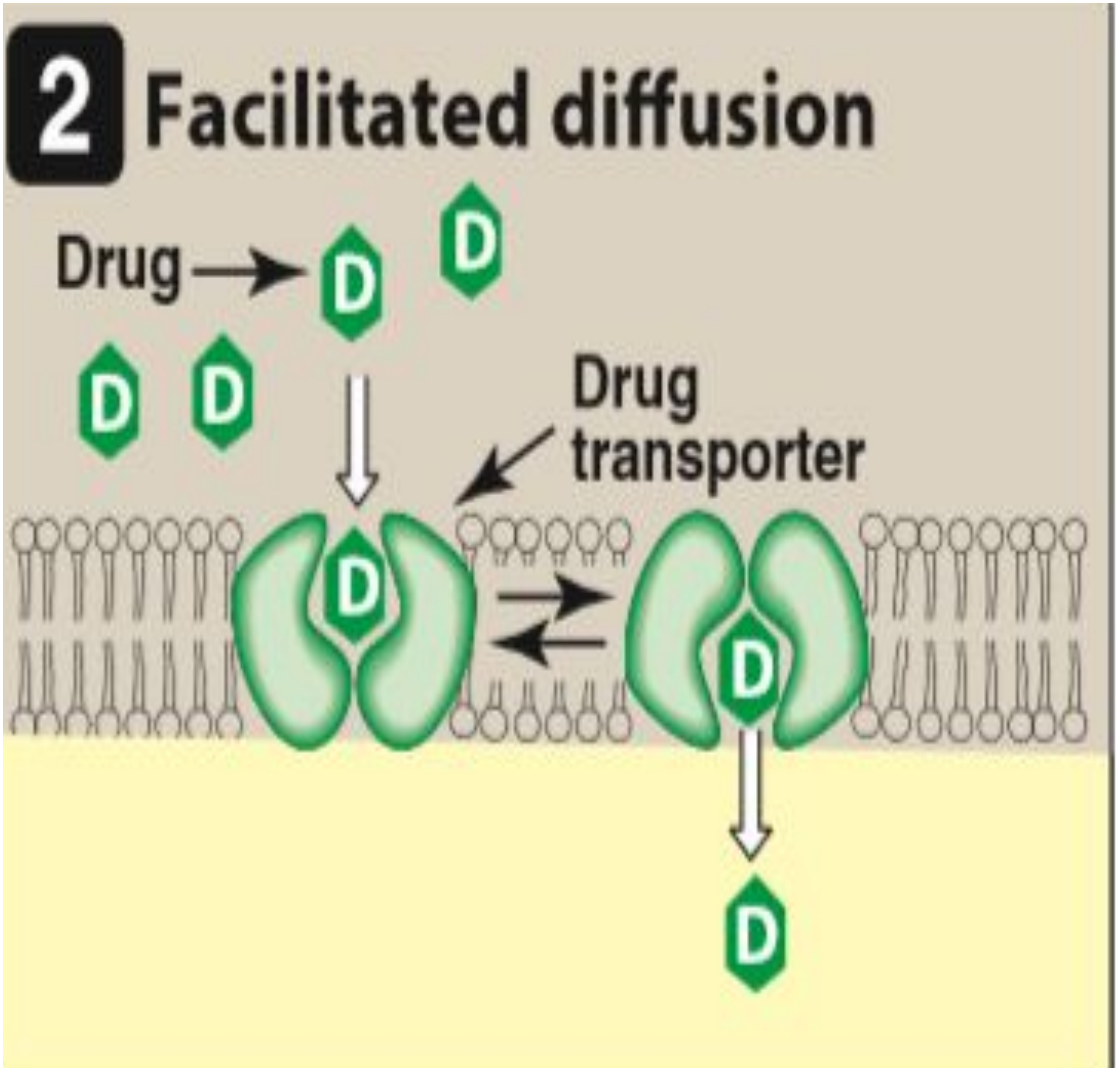


1 Passive diffusion
Passive diffusion of a water-soluble drug through an aqueous channel or pore
Passive diffusion of a lipid-soluble drug dissolved in a membrane
Drug
Drug
D
Extracellular space
Cell membrane
Cytosol

Mechanisms of absorption of drugs from the GI tract

2. ***Facilitated diffusion:***

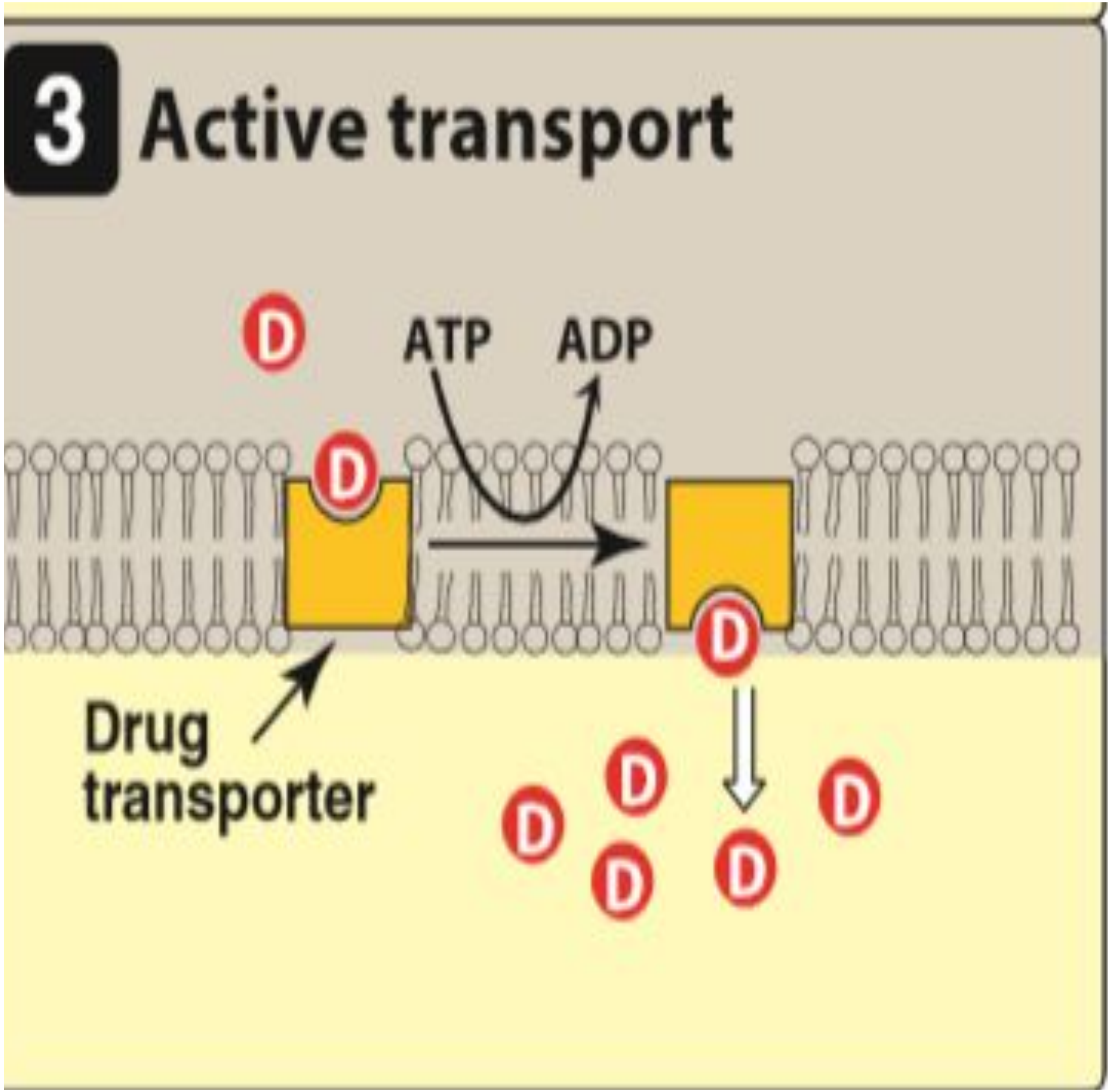
- There are specialized transmembrane carrier proteins that facilitate the passage of large molecules.
- Moving of drugs or endogenous molecules from an area of high concentration to an area of low concentration is known as facilitated diffusion.
- It does not require energy, can be saturated.
- May be inhibited by compounds that compete for the carrier.


2 Facilitated diffusion
Drug
D
Drug
transporter

Mechanisms of absorption of drugs from the GI tract

3. Active transport:

- From a region of low drug concentration to one of higher drug concentration.
- It involves specific carrier proteins.
- Energy-dependent active transport is driven by the hydrolysis of adenosine triphosphate.
- It is capable of moving drugs against a concentration gradient.
- The process is saturable.
- Active transport systems are selective and may be competitively inhibited by other cotransported substances.


3 Active transport
D
ATP
ADP
D
D
Drug
transporter
D
D
D
D
D
D

Mechanisms of absorption of drugs from the GI tract

4. Endocytosis and exocytosis:

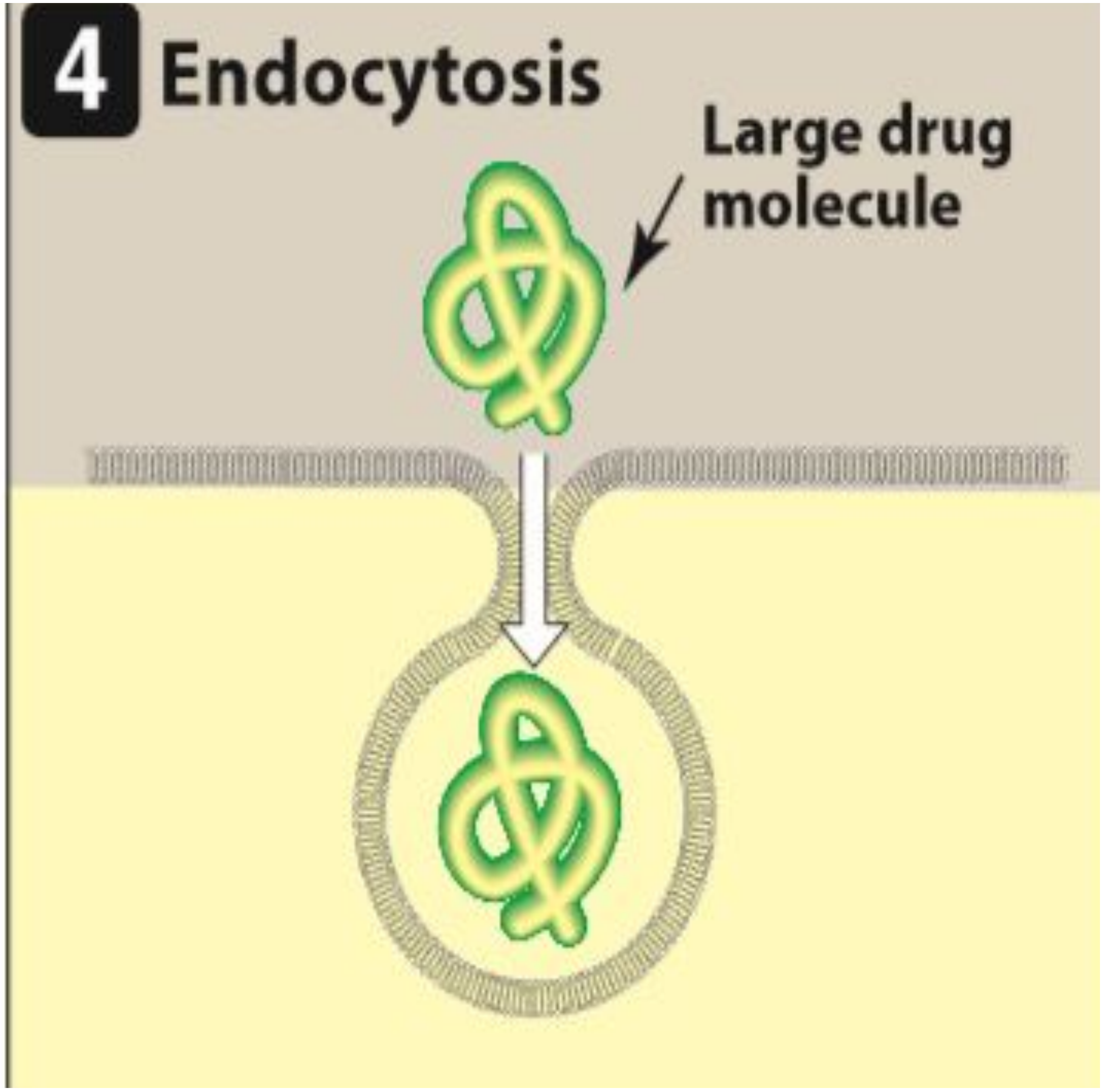
- Endocytosis involves engulfment of a drug by the cell membrane and transport into the cell by pinching off the drug filled vesicle.

For example Vitamin B12

- Exocytosis is the reverse of endocytosis.
- Many cells use exocytosis to secrete substances out of the cell through a similar process of vesicle formation.

For example Norepinephrine

This type of absorption is used to transport drugs of exceptionally large size across the cell membrane.


4 Endocytosis
Large drug
molecule

FACTORS EFFECTING ABSORPTION

- ***Effect of pH on drug absorption:***

Acidic drugs (HA) release a proton (H+) causing a charged anion (A−) to form

(PROTONATED) HA $\rightleftharpoons$ H+ + A- (DEPROTONATED)

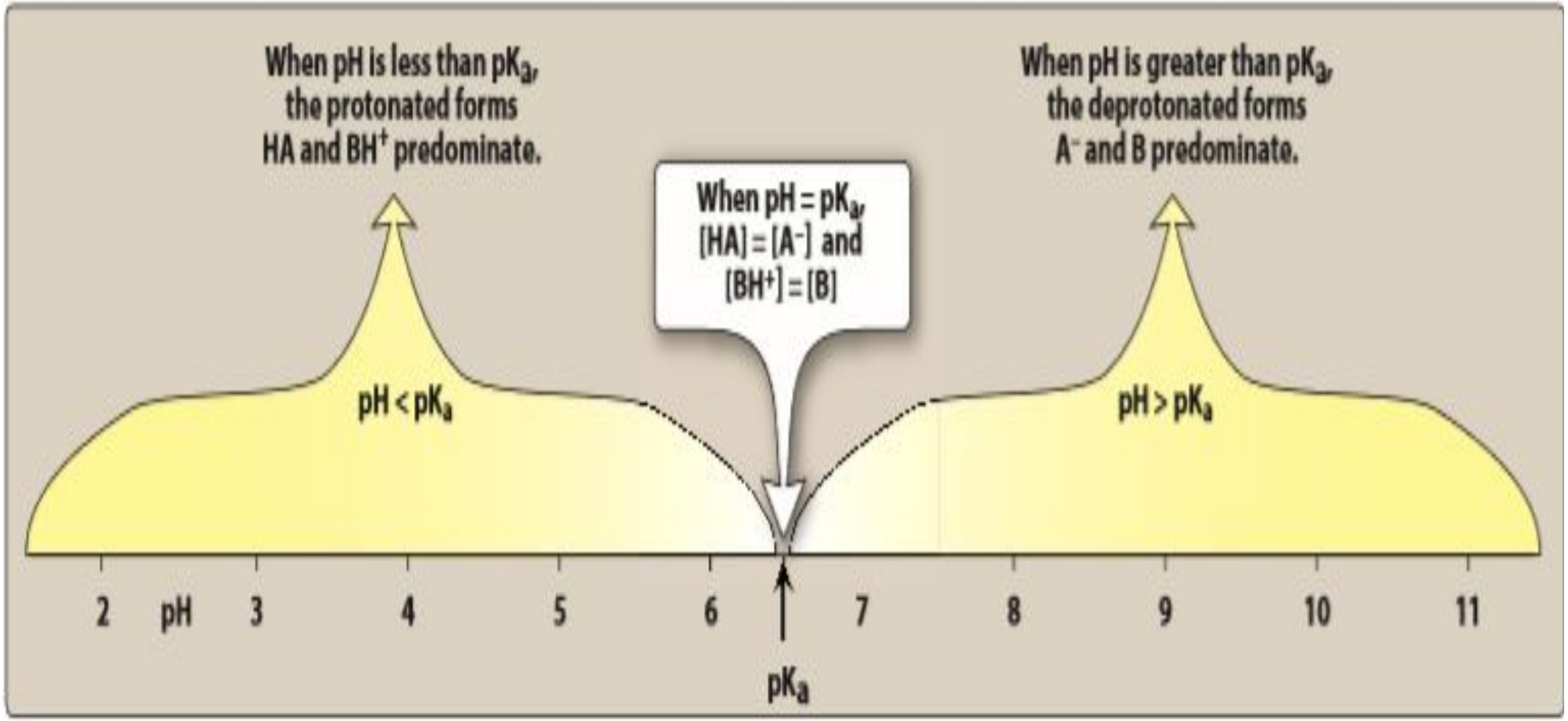
Weak bases (BH+) can also release an H+. However, the protonated form of basic drugs is usually charged, and loss of a proton produces the uncharged base (B)

(PROTONATED) BH+ B + H+ (DEPROTONATED)

FACTORS EFFECTING ABSORPTION

- A drug passes through membranes more readily if it is uncharged.
- Thus, for a weak acid, the uncharged, protonated HA can permeate through membranes, and A^- cannot.
- For a weak base, the uncharged form B penetrates through the cell membrane, but the protonated form BH^+ does not.
- ***Ionization constant (pKa)***

The pKa is a measure of the strength of the interaction of a compound with a proton. The lower the pKa of a drug, the more acidic it is. Conversely, the higher the pKa, the more basic is the drug.



When pH is less than pK$_a$, the protonated forms HA and BH$^+$ predominate.
When pH is greater than pK$_a$, the deprotonated forms A$^-$ and B predominate.
When pH = pK$_a$, [HA] = [A$^-$] and [BH$^+$] = [B]
pH < pK$_a$
pH > pK$_a$
2 pH 3 4 5 6 7 8 9 10 11
pK$_a$

Absorption (Summary)

Non-ionized drug

↓

More lipid soluble drug

↓

Diffuse across cell membranes more easily

FACTORS EFFECTING ABSORPTION

- ***Blood flow to the absorption site:***

The intestines receive much more blood flow than the stomach, so absorption from the intestine is favored over the stomach.

- ***Total surface area available for absorption:***

With a surface rich in brush borders containing microvilli, the intestine has a surface area about 1000-fold that of the stomach, making absorption of the drug across the intestine more efficient.

FACTORS EFFECTING ABSORPTION

- ***Contact time at the absorption surface:***

If a drug moves through the GI tract very quickly, as can happen with severe diarrhea, it is not well absorbed.

Conversely, anything that delays the transport of the drug from the stomach to the intestine delays the rate of absorption of the drug.

- ***Expression of P-glycoprotein:***

Areas of high expression, P-glycoprotein reduces drug absorption.

Bioavailability

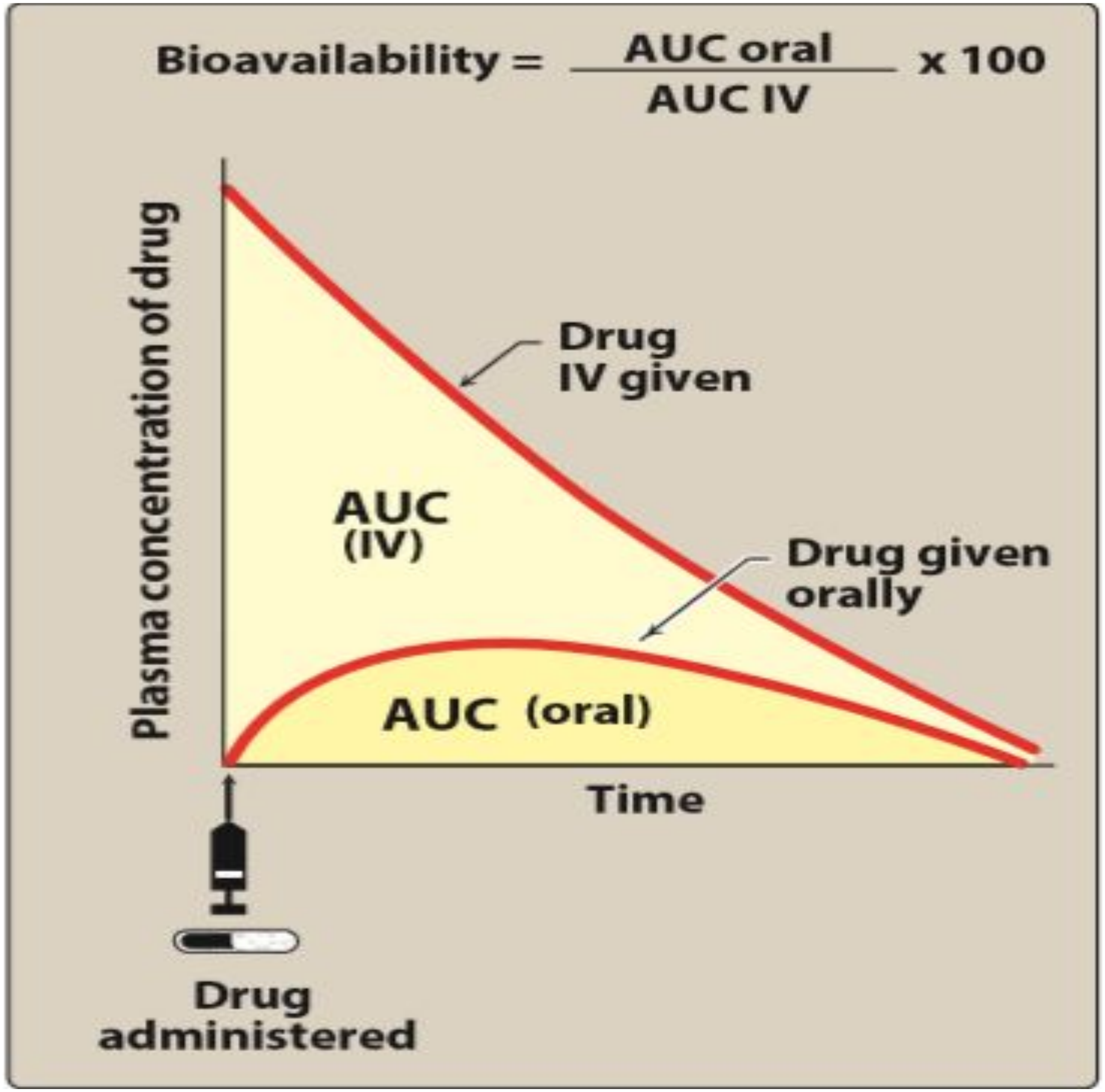
- Fraction of the administered dose reaching the systemic circulation

- Intravenous route

Absolute bioavailability

(F=1 or 100% bio-available)

DETERMINATION OF BIOAVILABILITY



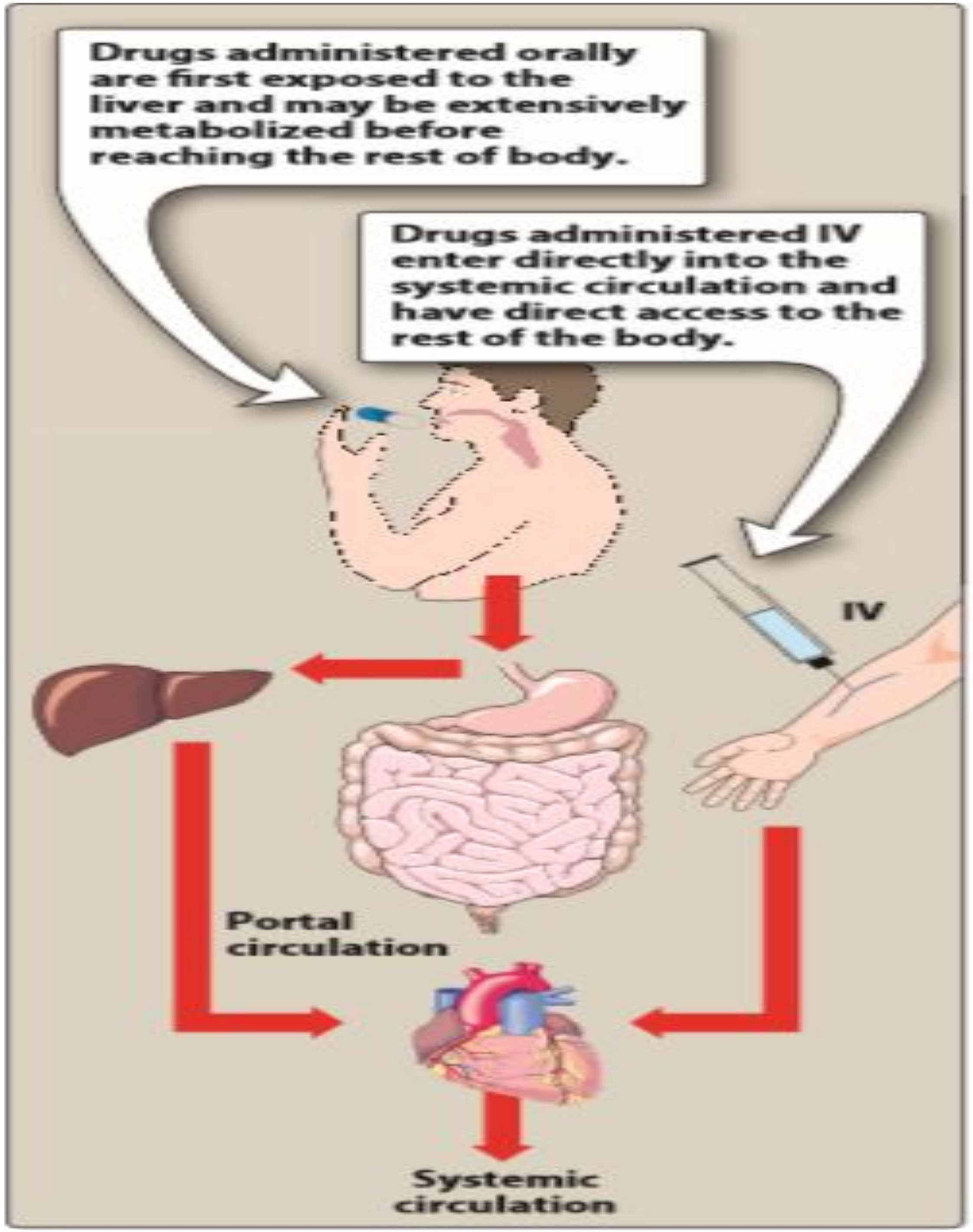
Factors that influence bioavailability

- ***First-pass hepatic metabolism***:

When a drug is absorbed from the GI tract,

it enters the portal circulation before

entering the systemic circulation.



Factors that influence bioavailability

- ***Solubility of the drug:***

Very hydrophilic drugs are poorly absorbed because of their inability to cross lipid-rich cell membranes.

Paradoxically, drugs that are extremely lipophilic are also poorly absorbed, because they are totally insoluble in aqueous body fluids and, therefore, cannot gain access to the surface of cells.

Factors that influence bioavailability

- ***Chemical instability:***

Some drugs, such as penicillin G, are unstable in the pH of the gastric contents. Others, such as insulin, are destroyed in the GI tract by degradative enzymes.

Factors that influence bioavailability

- ***Nature of the drug formulation:***

Drug absorption may be altered by factors unrelated to the chemistry of the drug. For example, particle size, salt form, crystal polymorphism, enteric coatings, and the presence of excipients can influence the ease of dissolution and, therefore, alter the rate of absorption.

THANKS